# TEDO AI Research Lab: A Permissively Licensed Desktop Workbench for Reproducible Machine Learning, Computer Vision and Quantum Simulation

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***Abstract*—Machine-learning research on a single workstation is usually spread over notebooks, scripts, terminals and hosted trackers, which makes experiments hard to repeat and leaves licensing and data-provenance questions open. We present TEDO AI Research Lab, an open-source desktop workbench built with Qt for Python in which an experiment is one declarative file that runs in a background worker process, is recorded in SQLite and MLflow, and leaves a run folder with a reproducibility snapshot from which it can be repeated. The workbench integrates image classification with PyTorch, eleven scikit-learn and XGBoost models, an interactive CNN explainer, a network designer, a dataset hub that checks licences at their sources, and a simulation laboratory for quantum circuits, wave packets and computer vision. Its newest module solves the Kohn–Sham equations self-consistently for electrons in a two-dimensional crystal lattice and renders the lattice potential, sampled electrons and averaged density as interactive three-dimensional views drawn without OpenGL. Every dependency and build tool is permissively licensed, a rule enforced by the tests. The physics engine reproduces the Madelung energies of the two-dimensional Wigner crystal to six digits and converges the default lattice in seven iterations and 0.6 s on a four-core CPU; 574 automated tests pass on Linux and Windows.**

***Index Terms*—experiment tracking, reproducibility, density functional theory, Kohn–Sham equations, scientific visualization, open-source software**

## I. Introduction

A machine-learning (ML) result depends on code, data, hyperparameters and the software environment; missing environment details and undocumented runs are a common reason results cannot be repeated [1]. Notebooks [2] and experiment trackers such as MLflow [3] address parts of this, but a researcher working alone still moves between many tools, and several popular vision libraries carry copyleft licences that restrict how derived work may be shared.

TEDO AI Research Lab keeps the whole cycle in one desktop application: choose a dataset and model, run, track, compare, explain and reproduce, on one machine with a 4 GB GPU. Its contributions are:

- an architecture in which every experiment is a strict YAML file run by a worker process that streams JSON events, recorded in SQLite and MLflow with a reproducibility snapshot;
- integrated learning laboratories and a dataset hub with a licence policy enforced in code;
- a simulation laboratory with a self-consistent plane-wave Kohn–Sham solver [4] for two-dimensional lattices and interactive 3-D views;
- validation of the physics against exact results and of the software by 574 tests on two operating systems.

## II. Related Work

MLflow [3] tracks parameters, metrics and artifacts, and Jupyter notebooks record computations [2]; the lab uses both inside a local workbench that also runs the experiments. Training builds on PyTorch [5], scikit-learn [6] and XGBoost [7], with SHAP values [8]. CNN Explainer [9] visualises a convolutional network in the browser; we port its views to a native engine that reproduces PyTorch's activations exactly. Codes such as Quantum ESPRESSO [10] solve the Kohn–Sham equations for real three-dimensional materials; our lattice model is a compact two-dimensional counterpart in which every step of the self-consistent loop can be inspected.

## III. System Architecture

The code is layered (Fig. 1): `core` holds contracts and engines without user-interface code, `labs` the domain logic, `plugins` the integrations (Kaggle, Hugging Face, Roboflow, Colab, MLflow) and `app` the Qt for Python (PySide6) interface with 29 pages. Tests enforce the import rules and a 400-line limit per module; the system has 286 modules and about 35 600 lines, with 7 700 more in tests.

### A. Experiments

A specification names a task, dataset, model, training settings and device; it is validated strictly and written in canonical form. Launching creates a run folder, and a job queue starts a worker subprocess that prints one JSON event per line and exits with 0, 1 or 2 for completed, failed or cancelled; cancellation is cooperative through standard input. Events are

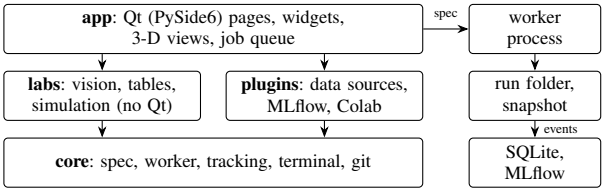



Fig. 1. Layers of the workbench and the life of an experiment.

recorded in SQLite with paths relative to the workspace, and the worker mirrors each run to MLflow without ever failing on a tracking error. A snapshot stores the Python version, platform, packages, git commit with uncommitted changes and a dataset fingerprint; *Reproduce* queues the same file as a child run and lists environment differences.

### B. Licences, Data and the Interface

Every dataset and model card carries a licence category checked at its source; downloads start only when the user asks and gated items are never downloaded. All dependencies and build tools must be MIT, Apache-2.0, BSD, ISC or PSF licensed, which the tests verify; the one exception is the LGPL Qt binding, for which no permissive alternative exists. The Windows installer is built with cx_Freeze. The interface never blocks: long work runs in worker processes or a thread pool, and every page is tested to fit half the window.

## IV. Laboratories and Simulation

### A. Learning Laboratories

The vision laboratory trains LeNet-5 [11], ResNet-18 [12], a small CNN and TinyVGG with mixed precision, checkpoints, resumption and early stopping; trained TinyVGG models open in the CNN explainer. The classical laboratory runs eleven models with cross-validation, search, ROC curves, permutation importance and SHAP, and a network designer writes the PyTorch code of a layer stack. The simulation laboratory builds quantum circuits of up to ten qubits, propagates wave packets by the split-step Fourier method [13], generates synthetic COCO-format detection data [14] and models a pinhole camera.

### B. Electrons in a Lattice

Ions of charge $Z$ sit on a square, triangular or honeycomb lattice a height $d$ above a plane of electrons, which feel $-Z/\sqrt{r^2+d^2}$ from each ion. In Kohn–Sham theory [4] the electrons are described by orbitals $\psi_i$ in an effective potential (Hartree atomic units),

$$\left[-\tfrac{1}{2}\nabla^2 + V_{\mathrm{eff}}(\mathbf{r})\right]\psi_i = \varepsilon_i \psi_i, \quad V_{\mathrm{eff}} = V_{\mathrm{ion}} + V_{\mathrm{H}} + V_{\mathrm{xc}}, \tag{1}$$

$$n(\mathbf{r}) = \sum_i f_i \,|\psi_i(\mathbf{r})|^2, \qquad f_i = \frac{2}{1 + e^{(\varepsilon_i - \mu)/k_B T}}, \tag{2}$$

where $V_{\mathrm{H}}$ is the Coulomb potential of $n$ itself and $V_{\mathrm{xc}}$ is the local density approximation of the two-dimensional electron gas: exact exchange and the correlation of Attaccalite *et al.* [15]. Orbitals are expanded in plane waves with $\tfrac{1}{2}|\mathbf{G}|^2 \le E_{\mathrm{cut}}$ at the $\Gamma$ point of the simulated cell and the Hamiltonian is diagonalised densely. Fermi smearing [16] fixes $\mu$, and Pulay mixing [17] of successive densities is repeated until $\int |n_{\mathrm{out}} - n_{\mathrm{in}}|\, dA/N < 10^{-6}$. The total energy adds an Ewald sum [18] for the ions and the energy $2\pi d N^2/A$ of the charge sheets.

The electrons on screen are samples of $n(\mathbf{r})/N$, not classical particles: walkers take Metropolis-adjusted Langevin steps [19], so their long-run distribution is the density, and a walker reaching another ion is drawn as a hop (Fig. 2).

TABLE I
Verification Against Exact or Independent Results

| Quantity | Lab | Reference |
|---|---|---|
| Madelung, square ($Er_s/N$, Ha) | $-1.100244$ | $-1.100244$ [20] |
| Madelung, triangular | $-1.106103$ | $-1.106103$ [20] |
| $\varepsilon_{\mathrm{c}}(r_s{=}1)$ (Ha) | $-0.11055$ | $-0.11009$ [21] |
| $\varepsilon_{\mathrm{c}}(r_s{=}10)$ (Ha) | $-0.03027$ | $-0.03045$ [21] |
| Sampler total variation | 0.014 | 0 |

TABLE II
Self-Consistent Runs ($4 \times 4$ Cells, $E_{\mathrm{cut}} = 8$ Ha)

| Lattice | $N$ | Waves | Iter. | Time (s) | $E/N$ (Ha) |
|---|---|---|---|---|---|
| Square | 16 | 509 | 7 | 0.60 | $-0.38247$ |
| Triangular | 16 | 435 | 7 | 0.40 | $-0.37713$ |
| Honeycomb | 32 | 379 | 7 | 0.33 | $-0.22322$ |
| Square, vacancy | 15 | 509 | 15 | 1.08 | $-0.38353$ |
| Square, $Z{=}2$, $d{=}0.6$ | 32 | 509 | 9 | 0.63 | $-1.00024$ |

### C. Three-Dimensional Views

Qt's own 3-D modules are GPL-licensed, so all 3-D views use Qt's 2-D painter: a perspective orbit camera, shaded quadrilaterals sorted far to near, and a cached image of the static scene with moving spheres drawn over it. The same camera draws a dataset's rows as a point cloud along three columns or principal components, coloured by class, in the dataset hub and the results.

## V. Validation

Table I compares the engine with independent results. The Ewald sum reproduces the Madelung energies of the two-dimensional Wigner crystal [20] for any splitting parameter; the correlation agrees with the fit of Tanatar and Ceperley [21] within 0.6 %; plane-wave eigenvalues are variational upper bounds within $10^{-5}$ Ha of exact diagonalisation on the full grid; and the walkers reproduce the density to a total variation distance of 0.014. For the default lattice the density change falls from 0.19 to $1.6 \times 10^{-7}$ in seven iterations, and raising $E_{\mathrm{cut}}$ from 8 to 20 Ha changes the energy per electron by $4 \times 10^{-6}$ Ha. Table II gives times on a four-core 2.8 GHz Xeon; $Z = 2$ with deeper wells fills the lowest band and opens a gap of 0.236 Ha.

The software is checked by 574 automated tests, run on every change on Ubuntu (Python 3.11 and 3.13) and Windows (Python 3.11); a smoke test opens all 29 pages and runs each

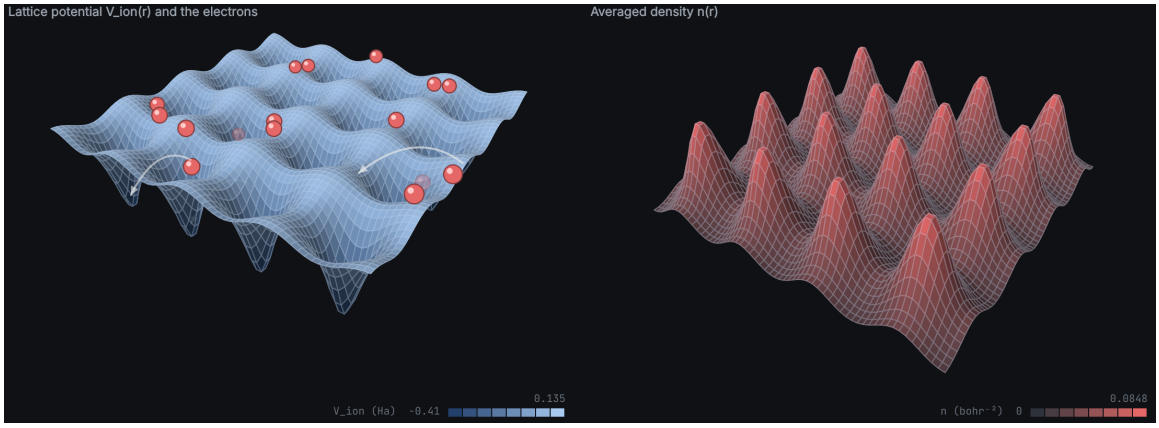



Fig. 2. The Lattice tool: (left) the ionic potential $V_{\text{ion}}$ of a $4 \times 4$ square lattice with sampled electrons and arrows marking hops between ions; (right) the self-consistent density $n(\mathbf{r})$. Both views are drawn with Qt's 2-D painter and turn together.

engine, and the Windows installer is built, installed silently and started in the same pipeline.

## VI. Discussion and Conclusion

The lattice model is deliberately small: two-dimensional, spin-unpolarised, local-density and sampled at the $\Gamma$ point only, with at most 1 600 plane waves; within these limits it exposes every step of the self-consistent loop. TEDO AI Research Lab shows that one permissively licensed desktop application can run, track, explain and reproduce ML experiments while hosting physics simulations verified against exact values. Planned work adds band structures and spin, 3-D views of learned features, and laboratories for audio and language.

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